Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

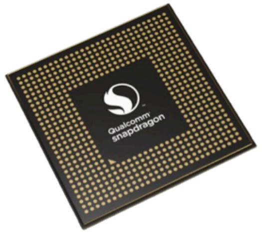
Software

Introduction to Computer Programming

Management and Artificial Intelligence

LUISS

How can a computer execute a program?

The **CPU** is responsible for executing the program code:

- It interprets the instructions one by one from the **RAM**
- The instructions must be written in **machine code**
- The machine code is:
    - **Low level**: relatively simple operations
    - **CPU specific**: different CPUs have different machine codes
    - **Binary format**: humans struggle to read it

Machine code: Intel/AMD vs ARM

A simple sequence of instructions to compute the sum of two integer numbers would look like this in machine code:

Intel/AMD 64-bit machine code

```
f3 0f 1e fa
55
48 89 e5
89 7d fc
89 75 f8
8b 55 fc
8b 45 f8
01 d0
5d
c3
```

ARM 64-bit machine code

```
ff 43 00 d1
e0 0f 00 b9
e8 0f 40 b9
e9 0f 40 b9
00 7d 09 1b
ff 43 00 91
c0 03 5f d6
```

- Each instruction is shown on a separate line for clarity, but stored contiguously in memory
- Machine code is binary (shown here in hexadecimal for readability)
- Intel/AMD processors share the same machine code format
- Other processors like ARM use entirely different machine code formats

From machine code to assembly code

To make machine code understandable for humans, we can translate it to assembly code:

- **Low level** [same as machine code]: relatively simple operations
- **CPU specific** [same as machine code]: different CPUs have different machine codes
- **Human readable** [differently from machine code]: CPU struggles to read it!

Intel/AMD 64-bit assembly	ARM 64-bit assembly

```
endbr64
push   %rbp
mov    %rsp,%rbp
mov    %edi,-0x4(%rbp)
mov    %esi,-0x8(%rbp)
mov    -0x4(%rbp),%edx
mov    -0x8(%rbp),%eax
add    %edx,%eax
pop    %rbp
ret
```

```
sub     sp, sp, #16
str     w0, [sp, #12]
ldr     w8, [sp, #12]
ldr     w9, [sp, #12]
mul     w0, w8, w9
add     sp, sp, #16
ret
```

Without getting into details, we can see that this code is doing some basic operations, such as addition, multiplication, copy/move of values from/to registers (which are different from CPU to CPU), etc.

NOTE:

- Translation from/to machine code to/from assembly code is a one-to-one non-ambiguous mapping.
- Translating from machine code to assembly code is called **disassembly**
- While the opposite is called **assembly**

Writing low level code is impractical

Major problems when working with machine/assembly code are:

1. Programmers have to write a lot of code to perform simple operations
2. Code is not portable across different hardware platforms
3. Programmers have to deal with hardware details
4. Programming in machine/assembly code is hard and error-prone

Yet, CPUs only understand and run machine code.

How can programmers survive?

From low-level to high-level programming

Most programmers use high-level languages, such as Python, JavaScript, or PHP, to write software.

However, the CPU does not understand high-level languages. Never!

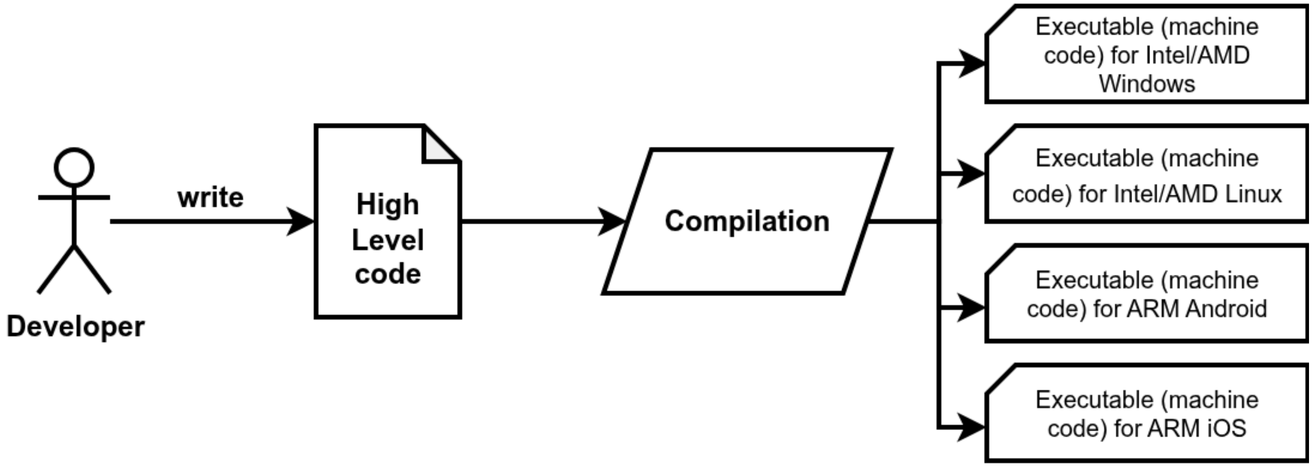
Code written in high-level languages must be translated into machine code. Two main strategies:

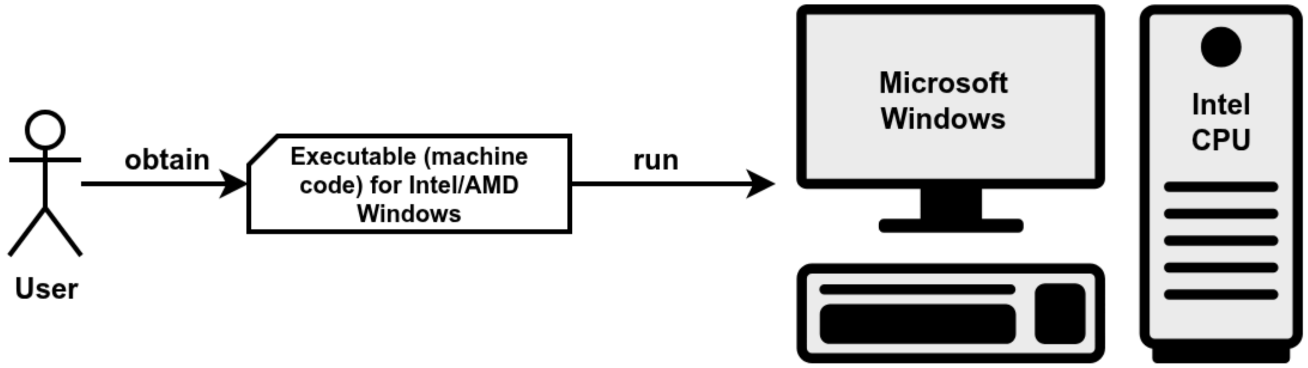
1. **Compilation**:
    - Compilers translate high-level code into machine code at *compile time*
    - Compile time is when the developer compiles the code (before it is run).
    - Machine code is packaged into an executable.
    - One executable for each platform (Operating Systems and CPUs).
    - Developers keep the high-level code, while users get the low-level code.

2. **Interpretation**:
    - Interpreters translate high-level code into machine code at *runtime*.
    - Running time means when the user runs the code.
    - The code is portable and platform-independent.
    - Developers distribute the high-level code, users use the interpreter to run the code.

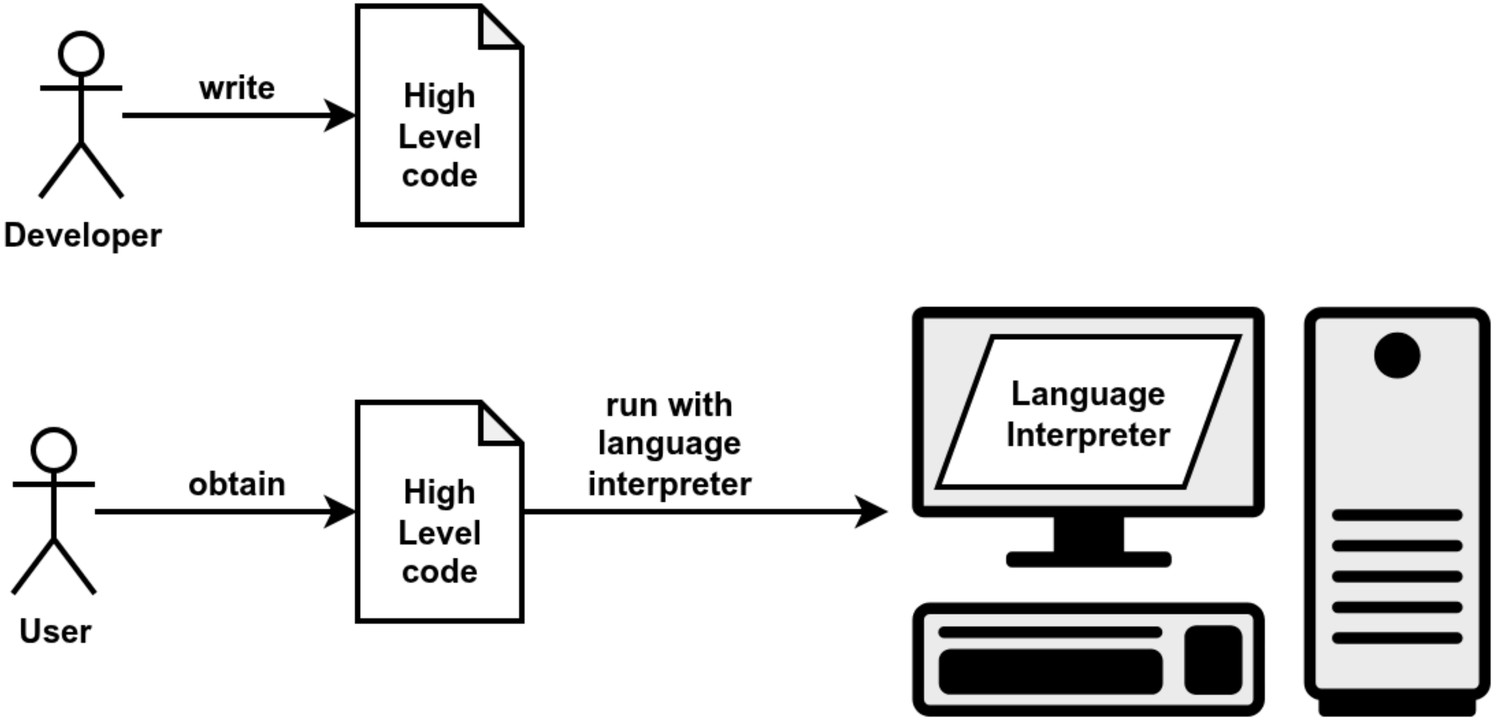
Compiled code is typically more efficient than interpreted code.

Compilation





Interpretation



Why do we need to learn a programming language?

Most everyday tasks require performing:

- repetitive…
- long sequences of…
- simple steps

This scenario is *exactly* where a computer excels!

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This scenario is *exactly* where a computer excels!

> *Computers are incredibly fast, accurate, and stupid. Human beings are incredibly slow, inaccurate, and brilliant. Together they are powerful beyond imagination.*
>
> *-- Albert Einstein*

Language: syntax and semantics

Programming languages have:

- **Syntax**: it defines which strings of characters and symbols are well-formed.
- **Semantics**: it defines the meaning of the well-formed strings.

The same is true even for natural languages (English, Italian, etc.):

- **Syntax**:
  - *"I are big"* is syntactically incorrect (*"I am big"* is correct).
  - *"The square circle is big"* is a well-formed sentence.
- **Semantics**:
  - *"The square circle is big"* is semantically incorrect (nonsensical because "square" and "circle" contradict).

An important difference between programming languages and natural languages is that programming languages are designed to be **unambiguous**, which means that there is no room for interpretation.

Programming language paradigms

Two major computing paradigms:

- **Imperative programming**:
  - we describe how the program must accomplish the task
  - language statements describe the steps (how to do), just like a **recipe**
  - close to the paradigm adopted by machine code
- **Declarative programming**:
  - we describe what to do (define the end goal), without saying how (no step-by-step)
  - language statements describe facts (what to do)
  - paradigm adopted, e.g., by Database programming languages (SQL)
  - distant from the paradigm adopted by machine code

Imperative programming is what we will see in this course.

NOTE: In many cases, code will be a mixture of both designs too, so it’s not always black-and-white.

Solving a task → Algorithm → Recipe

An algorithm features:

- a sequence of **simple steps**
- **flow of control** to specify when each step must be executed
- a way of determining when to **stop**.

...just like a recipe!

Even a plain 1 + 2 + 3 + 4 + 5 =? is an algorithm!

A slightly more complex example: Newton's method

The square root of x is a number y such that $y \cdot y = x$ ← **declarative**!

Let's see an **imperative** version:

1. Start with a guess g
2. If $g \cdot g$ is "close enough" to x:
    A. Let $y = g$
    B. Stop
3. Otherwise: a. Let $g \leftarrow \frac{1}{2}\left(g + \frac{x}{g}\right)$ b. Repeat from step #2

Square root of 16 via Newton's method:

g	$g \cdot g$	$\frac{1}{2}\left(g + \frac{x}{g}\right)$
3	$3 \cdot 3 = 9$	$\frac{1}{2}\left(3 + \frac{16}{3}\right) \simeq 4.17$
4.17	$4.17 \cdot 4.17 \simeq 17.39$	$\frac{1}{2}\left(4.17 + \frac{16}{4.17}\right) \simeq 4.0035$
4.0035	$4.0035 \cdot 4.0035 \simeq 16.028$	$\frac{1}{2}\left(4.0035 + \frac{16}{4.0035}\right) \simeq 4$

NOTE: As already pointed out, we cannot write an algorithm in our own natural language (e.g., English). To make an algorithm interpretable by a computer, we need to learn a **programming language**.

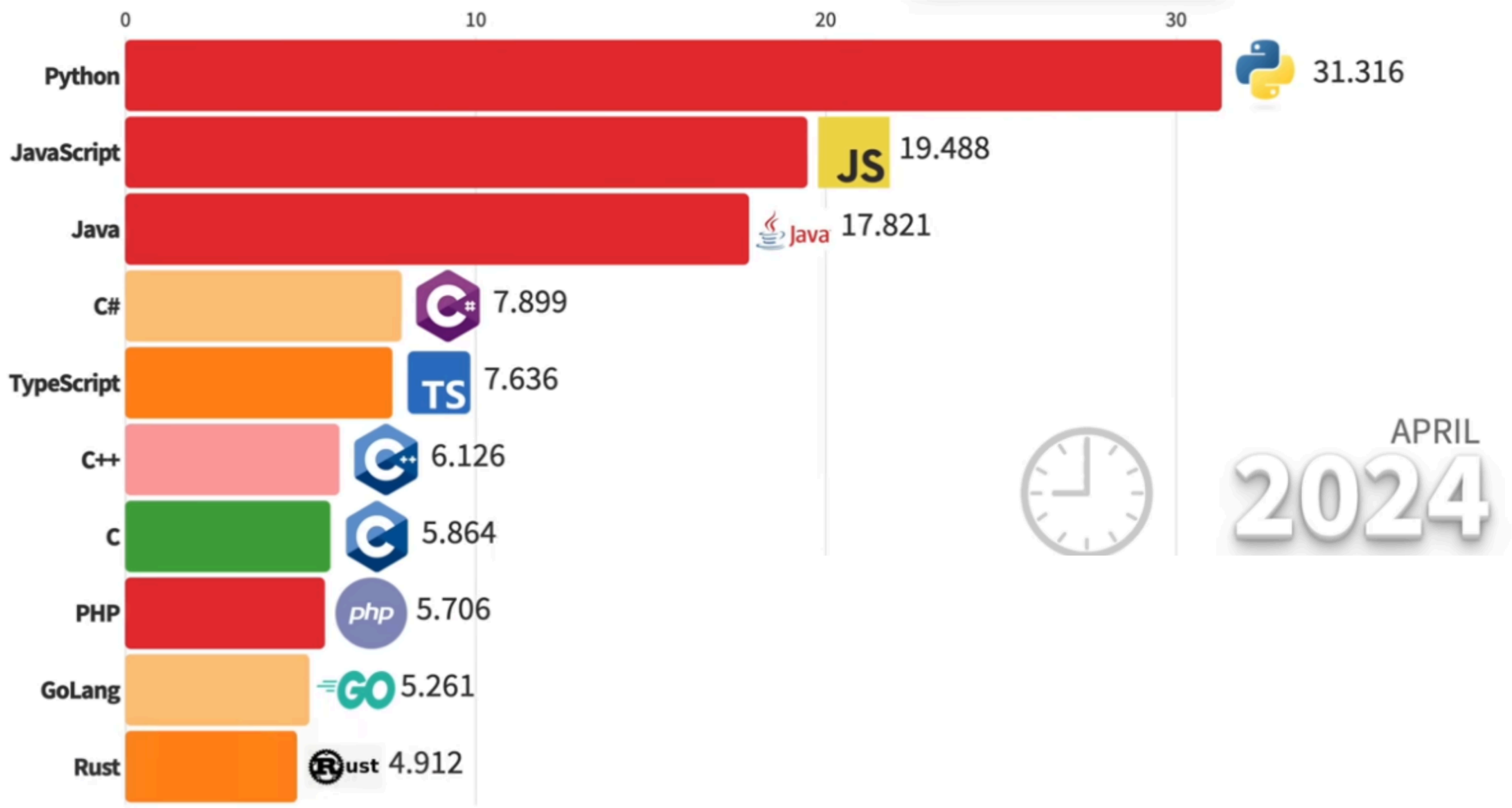
MANY different high-level languages

There are thousands of high-level languages:

- They have been designed with different goals in mind
- Some are general purpose, some are domain specific (e.g., web)
- Some give more control to the user, some hide more details
- Some are compiled, some are interpreted

In this course, we will focus on a single high-level language: ***but which should we choose?***

Popular programming languages



See the full video

Python

Benefits:

- **Easy to learn**: designed to be readable, quick to learn
- **High-level language**: abstracting away most low-level aspects of a computer
- **Large community and great support**: several libraries allow to build complex projects with just a few lines of code
- **Adequate for most tasks**: it shows its limits when performance is essential and low-level details play a role. You will hardly work on anything that cannot be implemented in Python

Python:

- is an **imperative programming language**
- with minor flavors of declarative (functional) programming traits.

Computers run complex software stacks

Python is nice and powerful, but computers are expected to run **MANY** programs at the same time:

- Browser to visit web pages (e.g., Chrome)
- Office editor to write documents
- File manager to organize files
- Messaging applications (e.g., Telegram)
- Video streaming client (e.g., Netflix)
- ...and way more!

Some of these may be written in Python, but others are written in other languages.

Hence, **a computer cannot just run a single Python program**. Moreover, computers are expected to:

- embrace the concept of user accounts
- provide access only after user authentication
- protect files across users and prevent hardware damage
- allow only admin users to install other programs
- handle network connections, power management

This is a **LOT** of **HARD** work: we need an **Operating System (OS)**.

Operating Systems

Modern computing systems are based on **operating systems**, which are software designed to allow the execution of applications (programs) on the available hardware.

The main tasks of an operating system are:

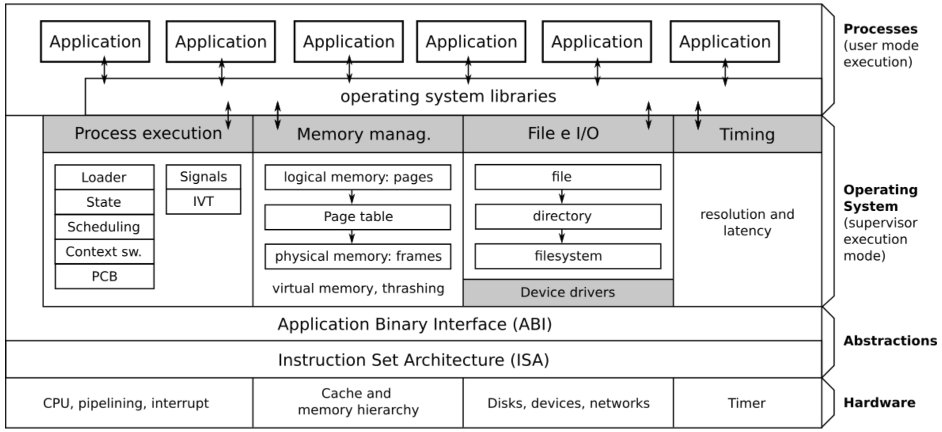
- **Process management** (creation, termination, CPU scheduling, etc.): any application (program) running on your OS is associated with a **process**.
- **Resource allocation** to processes (RAM, storage, network connections)
- **Time management** (clock, timers)
- **User management** for multi-user systems (access privileges to files, programs)
- **Inter-process communication** (locally or across networked systems)
- **Error handling** during process execution or system operations
- **Protection and security**, isolating processes to prevent interference with other processes, the OS, or hardware (preventing data theft or malicious/buggy operations)

Design goals of modern Operating Systems

Modern operating systems are designed with several key objectives:

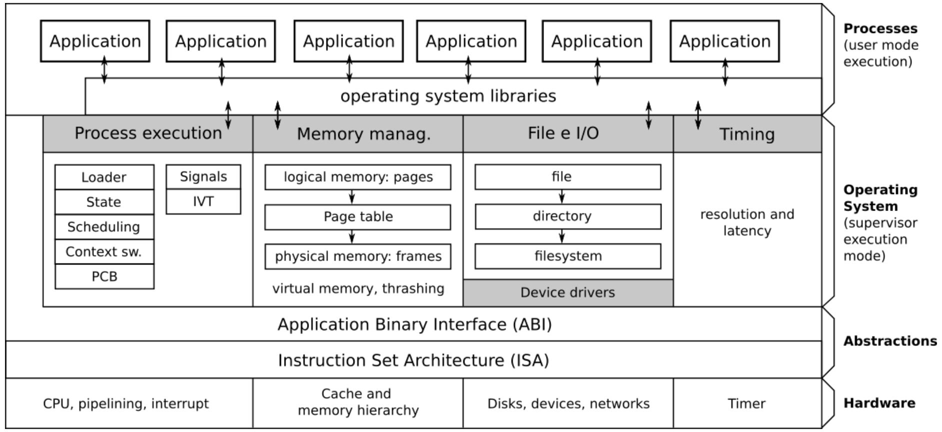
- **Multitasking**: Ability to run multiple applications simultaneously
- **Preemption**: Capability to take CPU control from one process and give it to another
- **Progress & fairness**: Regular CPU allocation to all processes to ensure computational progress
- **High throughput**: Maximizing the number of instructions executed per time unit
- **Low latency**: Minimizing response time for system operations (critical in real-time environments like automotive safety systems)

Operating System Overview



- The **hardware** is the physical part of the computer, containing CPU, RAM, disks and other devices.
- To program the hardware, each architecture defines the **Instruction Set Architecture (ISA)**, which defines machine code instruction syntax and semantics.
- The platform (operating system on a specific architecture, e.g., Windows x86) defines the **Application Binary Interface (ABI)**, which defines conventions (rules) on how different programs can interact, reuse code, and interface with the operating system.

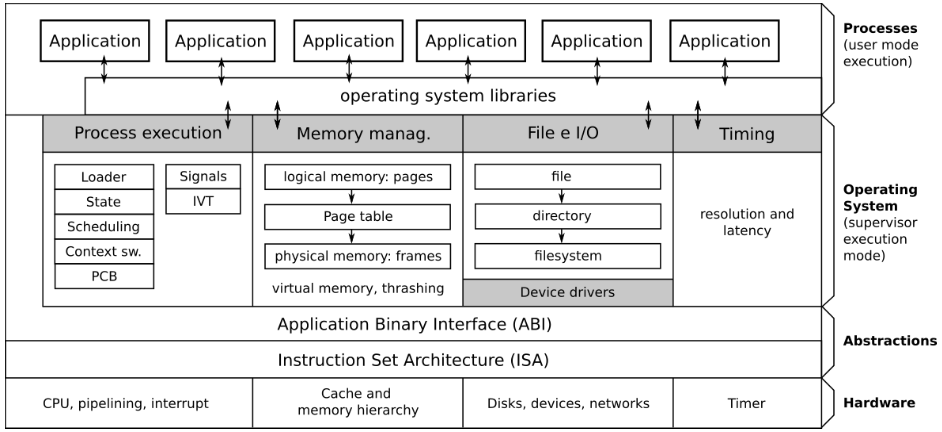
Operating System Overview (cont'd)



The **operating system**:

- has full control over the hardware:
  - it runs in **supervisor (kernel) execution mode**
  - it can execute **ALL** instructions from the ISA.
- controls which applications should run on the CPU cores at any time
- plays the role of intermediary between applications and the hardware
- manages four key aspects: **processes**, **memory management**, **file and I/O**, **time**

Operating System Overview (cont'd)



In **user execution mode**:

- Code can run only a limited (safe) subset of the ISA
- Privileged instructions must be delegated to the OS following the ABI
- Processes can be related to:
    - built-in applications provided by the OS: e.g., file manager
    - applications installed by the users: e.g., Chrome (web browser by Google)
- Applications use system libraries (subprograms for standard tasks, e.g., reading a file, make a network connection) to avoid reinventing the wheel when not necessary.

CPU Cores and Operating System Multitasking

- CPUs traditionally had only a single core (and many still have few cores)
- Yet we can run many applications simultaneously

How is this possible?

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How is this possible?

The Operating System creates the **illusion of parallel execution**:

- CPU time is divided into small time slots (milliseconds)
- Each application gets a turn to use the CPU
- When a time slot expires, the OS performs *context switching*
- Applications take turns using the CPU so quickly that users perceive them running simultaneously

This technique is called **time-sharing** or **preemptive multitasking** and it is the essential trait of all modern operating systems.

NOTE: If you have:

- 1 core without preemptive multitasking: up to 1 applications executing in parallel
- 1 core with preemptive multitasking: illusion of parallel execution of an infinite number of applications
- k cores without preemptive multitasking: up to k applications executing in parallel
- k cores with preemptive multitasking: illusion of parallel execution of an infinite number of applications, with up to k actually executing in parallel at any given time

Popular Operating Systems

Windows

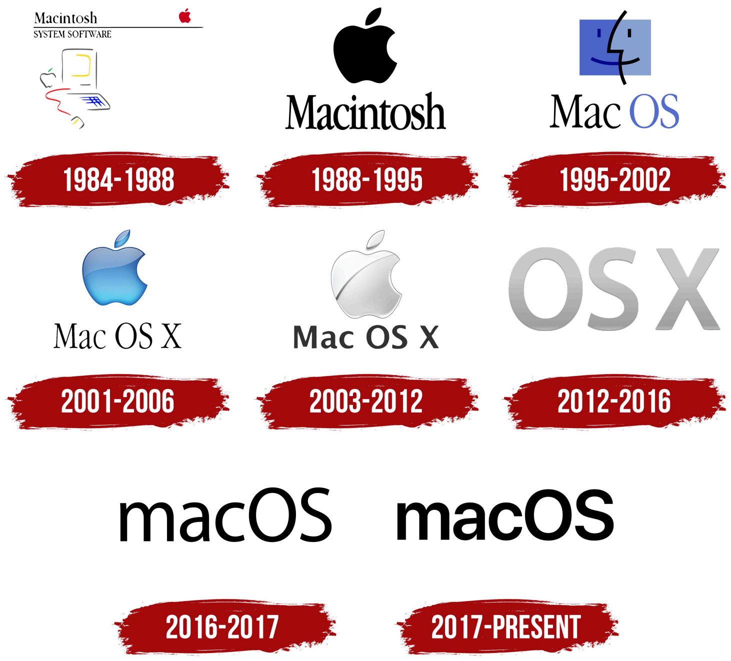
- Proprietary, closed-source OS developed by Microsoft for:
  - Intel/AMD CPUs
  - ARM (still early stage)
- Most widely used desktop OS globally
- Known for user-friendly interface and software compatibility

Windows 1 Windows 3.1 Windows 95 Windows XP Windows Vista Windows 7 Windows 8

Windows 10 Windows 11 Windows 12

macOS

- Proprietary, closed-source OS developed by Apple for:
  - PowerPC CPUs (before 2006)
  - Intel CPUs (before 2021)
  - Apple Silicon CPUs (now)
- Known for sleek design and stability
- Popular among creative professionals



GNU/Linux

- Combination of the Linux kernel and GNU userspace libraries
- Packaged by different *distributions* (Ubuntu, Fedora, Debian, Gentoo, Arch)
- **Open Source**: you can see and change its code!
- Powers most web servers and supercomputers (even Microsoft uses it in their cloud!)

Android

- Linux-based OS developed by Google
- Most popular mobile OS worldwide
- Open ecosystem with extensive app marketplace

iOS

- Proprietary, closed-source OS for Apple mobile devices
- Second most popular mobile OS worldwide
- Closed ecosystem with extensive app marketplace

Learn the OS Essentials

In the next lecture, we will learn about and work with essential operating system concepts.